3 Points Each

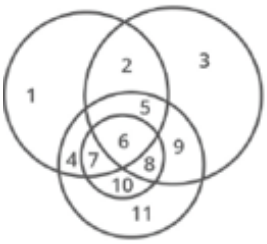
1999

1. Beata has two dolls, three apples, one chocolate bar, two oranges, five peaches, and one bike. How many pieces of fruit does Beata have?

(A) 3
(B) 5
(E) 3
(C) 10
(D) 18
(E) 21

2. What number is in the part that is common to four circles?

(A) 5
(B) 9
(C) 7
(D) 4
(E) 6



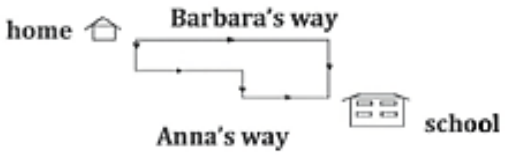
3. In how many places do we need to break a wooden stick in order to get 5 pieces?

(A) 3
(B) 4
(C) 5
(D) 6
(E) It depends on how long the stick is.

4. Karl is now 10 years old, and Alice is 3 years old. How many years from now will Karl be twice as old as Alice?

(A) 5
(B) 10
(C) 4
(D) 1
(E) 3

5. Anna and her sister Barbara go to the same school, but take two different ways to get to school. Whose way is shorter?



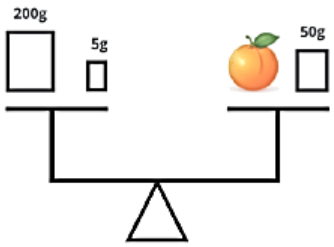
(A) Anna's way
(B) Barbara's way
(C) It depends on the distance to the school.
(D) Both ways have the same length.
(E) It is impossible to determine.

6. Our class has 30 students. The number of boys is four times the number of girls. How many girls are there in our class?

(A) 24
(B) 16
(C) 12
(D) 8
(E) 6

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7. How much does the orange weigh?



(A) 200 g
(B) 205 g
(C) 155 g
(D) 5 g
(E) It cannot be determined.

8. The cat says, "The length of my tail is 12 cm and half the length of my tail." How long is the cat's tail?

(A) 18 cm
(B) 24 cm
(C) 12 cm
(D) 9 cm
(E) 6 cm

4 Points Each

9. My mom's birthday is on a Sunday, and my dad's birthday is 55 days later. On what day of the week will my dad's birthday be?

(A) Sunday
(B) Monday
(C) Tuesday
(D) Thursday
(E) Saturday

10. Two basketball teams are playing against each other. The first team to win four games wins the tournament. There are no ties. What is the greatest number of games that could take place after which the winner will be known for sure?

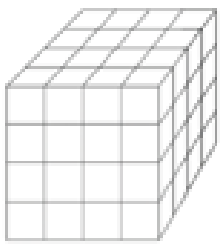
(A) 8
(B) 7
(C) 6
(D) 5
(E) 4

11. Instead of adding 27 to a certain number, John subtracted 27 from that number. What is the difference between John's result and the result he should have gotten?

(A) 27
(B) 0
(C) 54
(D) 100
(E) 3

12. A goldsmith decided to cut a golden cube with an edge of 4 cm into small cubes each with an edge of 1 cm. How many small cubes will he have?

(A) 64
(B) 48
(C) 32
(D) 16
(E) 12



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13. A pail filled with milk to the top weighs 25 kilograms, and a pail filled half-way weighs 13 kilograms. How much does an empty pail weigh?

(A) 2 kilograms
(B) 1½ kilogram
(C) 1 ½ kilograms
(D) 1 kilogram
(E) 2 1½ kilograms

14. In Grandma's pantry, there is a jar with 650 g of jam. Each day her grandson Tom eats 5 teaspoons of jam from the jar. Each teaspoon holds 6 g of jam. How much jam will there be left in the jar after 20 days?

(A) 50 g
(B) 530 g
(C) 550 g
(D) 1250 g
(E) The jar will be empty.

15. Each of the kangaroo's eleven children has eleven children, and each of them also has eleven children. How many great-grandchildren does the kangaroo have?

(A) 111
(B) 121
(C) 11211
(D) 1331
(E) 12321

16. What is the least possible number of children in the Kowalski family if each of the children has at least one brother and at least one sister?

(A) 1
(B) 2
(C) 3
(D) 4
(E) 5

5 Points Each

17. Peter opened a book and found that the sum of the page number on the left and the page number on the right is equal to 21. What is the product of the two page numbers?

(A) 121
(B) 100
(C) 420
(D) 110
(E) 426

18. Father Virgil is taking care of 143 children. Each day, each child gets half a liter of milk with breakfast. The milk from one cow is enough for 40 children. What is the least number of cows that Father Virgil needs to have?

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

19. A kangaroo wants to make a rectangular bedspread 1.5 m long and 1 m wide using square scraps which measure 10 cm x 10 cm. At every point where four squares meet she wants to place a fancy button. How many buttons will she need?

(A) 150
(B) 104
(C) 126
(D) 140
(E) 135

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20. Pinocchio's wooden nose is 3 cm long. Whenever Pinocchio lies, the length of his nose doubles. How long will his nose be after he tells 6 lies?

(A) 192 cm
(B) 67 cm
(C) 96 cm
(D) 18 cm
(E) 384 cm

21. In the yard there is an equal number of pigs, ducks, and chickens. Together, they have 144 legs. How many ducks are there in the yard?

(A) 18
(B) 21
(C) 35
(D) 42
(E) 43

22. One number was chosen from the numbers 51, 52, 53, 54, and 55, and the digit 0 was placed between the digits of that number. What is the difference between the new number and the number which was chosen?

(A) 500
(B) 50
(C) 550
(D) 450
(E) The difference depends on which number was chosen.

23. If Grandma gave each of her grandchildren 10 pieces of candy, there would not be any candy left over for one of the grandchildren. If she gave each one of them 8 pieces of candy, she would have 6 pieces of candy left. How many grandchildren does she have?

(A) 4
(B) 6
(C) 8
(D) 10
(E) 12

24. The figure shown rotates clockwise and makes one full rotation in one hour. Its position at 12:00 p.m. is shown in the picture to the right. What will it look like at 2:15 p.m.?



(A) 

(B) 

(C) 

(D) 

(E) 

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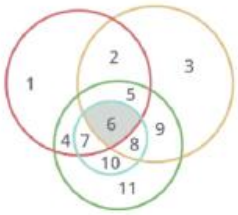
Answers key:

1. C
2. E
3. B
4. C
5. D
6. E
7. C
8. B
9. E
10. B
11. C
12. A
13. D
14. A
15. D
16. D
17. D
18. C
19. C
20. A
21. A
22. D
23. C
24. A

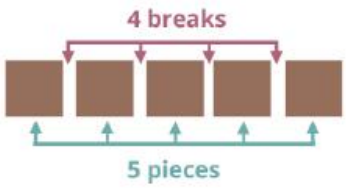
3 Point Solutions

1 (C) 10
Beata's fruits are: three apples, two oranges, and five peaches, so Beata has 3 + 2 + 5 = 10 pieces of fruit.

2 (E) 6
The picture has the circles colored to make them easier to see. The part common to all four circles is shaded.

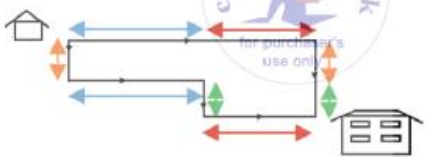


3 (B) 4
To get 5 pieces, you need to break the stick in only 4 places as shown in the picture below.



4 (C) 4
Karl is 10 – 3 = 7 years older than Alice. Because the difference of their ages is 7 years, he will be twice her age when she is 7, and he is 14. That will be 14 – 10 = 4 (or 7 – 3 = 4) years from now.

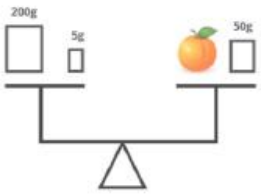
5 (D) Both ways have the same length.
The matching distances are marked by arrows of the same color.



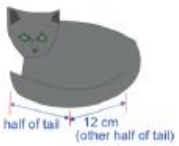
The matching shows that both ways have the same length.

6 (E) 6
We can split the boys into 4 groups, each of the same size as the group of girls. Thus, the class of 30 students consists of 5 equal groups and the number of girls is one-fifth of 30, which is 6.

7 (C) 155 g
Since the scales must balance,
5 g + 200 g = orange + 50 g
5 g + 150 g + 50 g = orange + 50 g
155 g = orange
The orange weighs 155 g.



8 (B) 24 cm
A whole is made up of two halves, so the 12 cm is half the length the tail. Hence, the tail is 12 cm + 12 cm = 24 cm long.



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4 Point Solutions

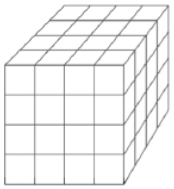
9 **(E) Saturday**
It is also a Sunday 8 weeks after mom's birthday. 8 weeks have 8 × 7 = 56 days and 56 − 1 = 55, so dad's birthday is one day before Sunday, which is Saturday.

10 **(B) 7**
6 games are not always enough since each team could win exactly 3 games. With 7 games one team must have at least 4 wins and the other team less than 4 wins.

11 **(C) 54**
Subtracting instead of adding is moving in the opposite direction on a number line. Instead moving to the right John moved 27 units to the left. The distance between John's result and the result he should have gotten is 27 + 27 = 54.



12 **(A) 64**
Look at the top layer of the big cube. It is made up of 4 × 4 small cubes since 4 cm = 4 × 1 cm. There are 3 other layers below the top layer, so there are altogether 4 layers of 4 × 4 small cubes. The big cube contains 4 × 4 × 4 = 64 small cubes.



13 **(D) 1 kilogram**
The pail plus half of the milk weighs 13 kilograms. If we add the other half of the milk to the pail, then the pail is filled with milk to the top and the total weight is 25 kilograms. Hence, half of the milk weighs 25 kilograms − 13 kilograms = 12 kilograms. All the milk weighs 2 × 12 kilograms = 24 kilograms, so the pail weighs 25 kilograms − 24 kilograms = 1 kilogram.

14 **(A) 50 g**
Tom eats 5 × 6 g = 30 g of jam from the jar each day. After 20 days he will eat 20 × 30 g = 600 g of jam, so 650 g − 600 g = 50 g of jam will be left in the jar.

15 **(D) 1331**
The kangaroo has 11 children, and 11 × 11 = (10 + 1) × 11 = 10 × 11 + 11 = 110 + 11 = 121 grandchildren. Each grandchild has 11 children, so the kangaroo has 11 × 121 = (10 + 1) × 121 = 10 × 121 + 121 = 1210 + 121 = 1331 great-grandchildren.

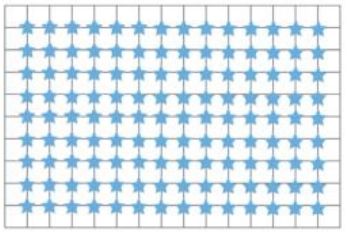
16 **(D) 4**
Each Kowalski child has a sister and a brother, so there is a girl and a boy in the Kowalski family. The boy has a brother, so there are at least two boys in the family. The girl has a sister, so there are at least two girls in the family. Thus, the least possible number of children in the family is 4.

5 Point Solutions

17 **(D) 110**
The page numbers are 10 and 11 since they are consecutive numbers that add up to 21.The product of the two page numbers is 10 × 11 = 110.

18 **(C) 4**
143 = 40 + 40 + 40 + 23. 3 cows can feed 40 + 40 + 40 children and we need at least one 1 more cow to feed the other 23 children, so Father Virgil needs at least 4 cows.

19 **(C) 126**



1 m = 10 × 10 cm and 1.5 m = 15 × 10 cm, so the bedspread is a rectangle made out of 10 × 15 square scraps. There are 9 rows of buttons (9 = 10 − 1) and there are 14 buttons (14 = 15 − 1) in each row, so the number of all buttons needed is 9 × 14 = 90 + 36 = 126.

20 **(A) 192 cm**
The length of Pinocchio's nose is:

2 × 3 cm after the 1st lie,
2 × 2 × 3 cm after the 2nd lie,
2 × 2 × 2 × 3 cm after the 3rd lie,
2 × 2 × 2 × 2 × 3 cm after the 4th lie,
2 × 2 × 2 × 2 × 2 × 3 cm after the 5th lie, and
2 × 2 × 2 × 2 × 2 × 2 × 3 cm after the 6th lie.

Consecutive products of 2 are: 2, 4, 8, 16, 32, and 64, so after 6 lies Pinocchio's nose is 64 × 3 cm = 192 cm long.

21 **(A) 18**
1 pig has as many legs as a duck and a chicken together, so 72 legs (half of the 144 legs) belong to the pigs. Each pig has 4 legs, so there are 18 pigs since 4 × 18 = 72. Hence, there are 18 ducks in the yard since the number of ducks is the same as the number of pigs.

22 **(D) 450**
The differences are: 501− 51 = 450, 502 − 52 = 450, 503 − 53 = 450, 504 − 54 = 450, and 505 − 55 = 450. Whichever number was chosen, the difference is always 450.

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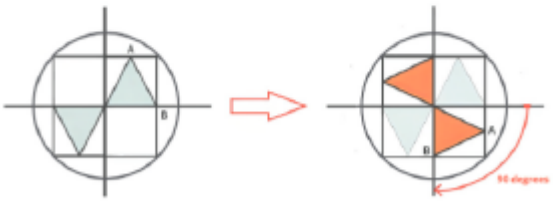
23 (C) 8

All the grandchildren except one received 10 pieces of candy each. After that Grandma wants each grandchild to have the candy. She can do it if every grandchild has 8 pieces of candy and there are still 6 pieces left over. Every grandchild with 10 pieces of candy must give up 2 pieces of candy, so Grandma can collect 8 pieces for the one grandchild without candy and keep 6 extra pieces. Together she would collect 8 + 6 = 14 pieces, which means that there were 7 grandchildren who originally had 10 pieces each, because 2 × 7 = 14. Including the grandchild originally without candy, she has 8 grandchildren.

24 (A)



At 2:00 p.m. the figure looks exactly the same as at 12:00 p.m. since any number of full rotations brings it back to how it was at first. Now, we need to see how much it rotates in the 15 minutes after 2 o'clock. Since 15 minutes is ¼ of an hour, the figure will rotate ¼ of a full circle, or 90°, clockwise. The figures below have lines added to help visualize the rotation better. The orange figure in the second picture shows the figure after the rotation.



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